

1. Key Resources

Many of these are already detailed in the syllabus. However, there are more that can be of service.

From the syllabus:

- ‘Daggity’: Daggity is a free online tool to help visualize confounding relationships. This can be used online or downloaded to your computer. You can interface here: <https://www.dagitty.net/>.
- *Kaggle*: Kaggle is an open access data repository, where you can find a plethora of different datasets. This is a great resource for exploring further data and honing your data science skills. You can find the repository here: <https://www.kaggle.com/datasets>.
- Whickam H, Cetinkaya-Rundel M, Grolemund G (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data* (2nd Edition). O’Reiley. <https://r4ds.hadley.nz/>
- Hernán MA, Robins JM (2020). *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>
- UCLA Regression Tutorial: [https://stats.oarc.ucla.edu/wp-content/uploads/2022/11/R_reg_2022.html#\(1\)](https://stats.oarc.ucla.edu/wp-content/uploads/2022/11/R_reg_2022.html#(1))

Further resources:

- Stack Overflow: This is an online community where people post questions and problems to message boards, and people respond and try to be helpful in answering these questions or solving the problems. There is almost always code you can copy and paste with great descriptions of what is happening.
- R Cookbook: This online book (and can also buy a hard copy), provides code solutions with explanations for many early code user issues. Really good reference for quick solutions and getting your code up to speed.
 - Website for book: <https://rc2e.com/>
- R Graphics Cookbook: This online book (and can also buy a hard copy), provides code solutions with explanations for many different data visualization. Really good reference for quick solutions and getting your code up to speed concerning visualizations.
 - Website for book: <https://r-graphics.org/>

2. R Packages

dplyr: For manipulating data. <https://www.rdocumentation.org/packages/dplyr/versions/1.0.10>

ggplot2: For data visualization. <https://ggplot2.tidyverse.org/>

naniar: For missing data assessment. <https://naniar.njtierney.com/>

stringr: For manipulating string data. <https://stringr.tidyverse.org/>

lubridate: For manipulating date data. <https://lubridate.tidyverse.org/>

janitor: For cleaning data. <https://www.rdocumentation.org/packages/janitor/versions/2.2.1>

Epi: Summarizes regression outputs. <https://www.rdocumentation.org/packages/Epi/versions/2.63>

car: Assesses regression diagnostics. <https://www.rdocumentation.org/packages/car/versions/3.1-5>

visdat: Assist ggplot2 package for data visualization.
<https://www.rdocumentation.org/packages/visdat/versions/0.6.0>

pmsampsize: Calculates sample sizes for prediction model development. <https://cran.r-project.org/web/packages/pmsampsize/index.html>

pROC: Calculates area under the curve for predictions.
<https://www.rdocumentation.org/packages/pROC/versions/1.19.0.1>

CalibrationCurves: Visualizes calibration plots for predictions. <https://cran.r-project.org/web/packages/CalibrationCurves/index.html>

Hmisc: Helper package for prediction (will not use any direct functions):
<https://www.rdocumentation.org/packages/Hmisc/versions/5.2-4>

rms: Helper package for prediction (will not use any direct functions):
<https://www.rdocumentation.org/packages/rms/versions/8.1-1>

3. Basic Key Commands

PC

Ctrl + enter : Run a line of code

Ctrl + Alt + R : Run the entire script of code

Ctrl + L : Clear the execute code window

Ctrl + shift + M : 'Pipe' -- %>%

Mac

Command + return : Run a line of code

Option + Command + R : Run the entire script of code

Command + L : Clear the execute window

Command + shift + M : 'Pipe' -- %>%

4. Pre-Course

Downloading and opening R and R Studio:	https://www.r-project.org/ https://posit.co/download/rstudio-desktop/
Installing R Tools:	https://cran.r-project.org/bin/windows/Rtools/
install.packages() :	R function to install packages.
Downloading data from Kaggle:	https://www.kaggle.com/datasets .
read.csv(filepath) :	R function to upload data to R.
library(RPackage) :	Activate a library to use in R
%>% :	‘Pipe’ to connect multiple functions
Tidy data:	Standardized way to order data, where each column is a variable and each row is an observation
R Studio Cloud Based – Posit:	www.posit.cloud

5. Joining & Transforming Data

Wide Data: Each subject has one row. Different measurement time points or conditions are separate columns

Long Data: Each row is a single observation. Each time point or condition is a different row

filter() : function: changes which rows are present without changing their order. Dplyr package.

arrange() : changes the order of the rows without changing which are present. Dplyr package.

distinct() : gives on exact match per row. Dplyr package.

%in% : Keeps rows where the variable equals one of the values on the right. Dplyr package.

mutate() : Creates new columns from other columns. Dplyr package.

select() : Changes which columns are present. Dplyr package.

rename() : Changes the names of the columns. Dplyr package.

relocate() : Changes the positions of the columns. Dplyr package.

clean_names() : Lower caps and cleans all variable names to computer readable format. Janitor package.

group_by() : Groups by different logic (left v right; male v female). Dplyr package.

summarize() : Aggregates by different commands (mean, median, etc.). Dplyr package.

slice() : Extract different rows from each group. Dplyr package.

left_join() :The data frame on the left will have the same # rows. Dplyr package.

inner_join() : Will only join data that is in BOTH. Dplyr package.

right_join() : The data frame on the right will have the same # rows. Dplyr package.

full_join() : All rows from BOTH data frames are included. Dplyr package.

semi_join() : Keeps only rows that are in the left with a match in right. Dplyr package.

anti_join() : Row that are NOT in BOTH data frames are included. Dplyr package.

6. Open Science and Data Protection

FAIR Principles for Open Science

F — Findable

Data and metadata should be easy for both humans and computers to discover.

A — Accessible

Once found, data can be retrieved using well-defined procedures.

I — Interoperable

Data can be combined with other data and used in different systems.

R — Reusable

Data can be reused for future research and new purposes.

De-Identification: Basic removing names and birthdates. Creating a unique ID

Pseudonymization: Renaming patients by an independent third party to improve patient data protection.

Data Caretaker: “Gate Keeper.” Limited disclosure of personal health information under a data caretaker that limits the scope which the data may be used or released.

Federated Data System: Database systems that are autonomous, where data are not replicated or copied from the original database.

Data Dictionary: Dictionary of each variable in a dataset, including definition, units, and any transformations performed.

Four eyes principle: All code should be checked by at least one other person ‘four eyes’ before depositing code.

set.seed() : ‘Setting seed’ functions allows for reproducibility of a randomized process.

anonymize(, .algo = "crc32") : Creates a random ID. Anonymizer package.

sample(variable, #, replace = FALSE) : Function that randomly samples from a variable. Can detail how many samples (#), and if you do or do not want to have replacement for sampling.

7. Cleaning Data

View() : Views the entire dataset in tabular 'Excel' format. Dplyr package.

head() : Provides the first 10 rows of the dataset. Dplyr package.

tail() : Provides the last 10 rows of the dataset. Dplyr package.

names() : Provides a list of every variable name within the dataset. Base R.

str() : Provides the structure of each variables within the dataset. Base R.

hist() : Creates a histogram of a variable. Base R.

count() : Counts the number of observations of a variable. Dplyr package.

str_trim : Trims the spaces on both side of a string that are 'empty'. Stringr package.

sapply : Wrapper function that applies a function on each row and return a vector or matrix. Base R.

df <- df[df\$a != 3,] : Deleting a specific cell data point, detailing the column and logic to find that cell data point. Base R.

8. Data Visualization

All of these functions are from the ggplot2 package.

ggplot(, aes()) : Base ggplot2 function, includes dataframe, and X and Y at a minimum.

+ **geom_point()** : Addition to ggplot function, to create a point for each piece of data.

+ **geom_smooth()** : Addition to ggplot function, to create a smooth line for the data.

+ **xlab()** : Addition to ggplot function, x-axis lab title function.

+ **ylab()** : Addition to ggplot function, y-axis lab title function.

+ **ggtitle()** : Addition to ggplot function, overall figure title function.

+ **labs()** : Addition to ggplot function, legend title function.

+ **scale_color_discrete()** : Addition to ggplot function, name each color level within the legend.

+ **geom_jitter()** : Addition to ggplot function, 'jitters' data points.

+ **facet_wrap()** : Addition to ggplot function, creates equal units figure side by side.

9. Missing Data

Missing Completely at Random (MCAR): Data is missing without a pattern.

Missing at Random (MAR): Data is missing with a pattern, but it is explained by the data at hand within the dataset.

Missing Not at Random (MNAR): Data is missing with a pattern, but is cannot be explained by the data at hand within he dataset.

sum(is.na()) : Base R function that gives the total number of missing data points within a dataset. Base R.

dim() : Base R function that provides the dimensions of a data set columns and rows. Base R.

miss_var_summary() : Provides the number and percent missing per variable in a dataset. Naniar package.

vis_dat() : Provides missing visuals of all data points within a dataset. Naniar package.

geom_miss_point() : Part of ggplot2 function, highlights missing data points. Ggplot2 & Naniar packages.

vis_miss() : Provides visual of missing data points by cluster. Naniar package.

gg_miss_upset() : Shows patterns of missingness by variable clusters. Ggplot2 & Naniar Packages.

gg_miss_var() : Shows cumulative missingness by variable. Ggplot2 & Naniar Packages.

bind_shadow() : Part of the shadow matrix, which provides each missing point as as new data point to compare against data that is observed. Naniar package.

10. Epidemiology

Epidemiology: The scientific study of the distribution, patterns, and determinants of health-related states or events (including diseases) in specified populations.

Exposure: Time at risk.

Injury: Physical damage to the body's tissues as a consequence of energy transfer and develops over a short period of time.

Recurrent Injury:

Subsequent Injury:

International Classification of Diseases: Classification of diseases by the World Health Organization

Occupational Injury & Illness Classification: Classification of injury and illness by the US Dept of Labor.

The Orchard Sports Injury Classification System (OSIICS): Classification of injury by body part for sport.

PICO: Population; Intervention; Comparison; Outcome.

Confounding: A systematic error in epidemiology where a third variable (confounder) distorts the true relationship between an exposure and an outcome, causing an apparent, yet false, association.

Person Time: The number of exposures for all involved patients.

Risk: Average probability of incurring the disease per patient.

Censor: Patient occurrence of disease does not happen during observation.

Right censor: Follow up ends without disease.

Left Censor: Disease happens before observation.

Incidence: The number of cases of disease with their onset during a prescribed period of time.

Rate: The number of cases of disease with their onset per unit of person time.

Prevalence: The number of cases with a disease or attribute (i.e., injury) in a specific population.

min() : Function that gives the minimum number in a variable. Base R.

max() : Function that gives the maximum number in a variable. Base R.

sum() : Function that sums the total numbers in a variable. Base R.

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round() : Function that rounds a number to prescribed number of digits. Base R.

11. Confounding

Confounding: Alternative more lay definition: The relationship between an independent variable (exposure) and a dependent variable (outcome) is distorted by the presence of another variable.

Association: Describe.

Predict: Forecast into the future.

Causal: Cause-effect.

Three Stages of Identifying Confounding: 1) Univariable Analyses; 2) Add Potential Covariates; 3) Externally Replicate and/or Evaluate in Different Scenarios.

Creating Confounding Structural Relationships: 1) Search & synthesize the literature; 2) Evaluate biological plausibility; 3) Interview experts (if necessary); 4) Create a structural diagram to explicitly detail this relationship.

Diagonal Acyclic Graph (DAG): Visually representation of causal relationships.

- **Directed:** The edges (arrows) between nodes (variables) have a direction, indicating a causal or dependency relationship
- **Acyclic:** There are no loops or cycles — you cannot start at one node and follow a path that leads back to it.

Common Cause: To both the exposure & outcome.

Collider: Caused by two or more other variables. Conditioning on the collider introduces bias into the analyses.

Mediator: A part of the causal pathway. Is caused by the exposure and influences the outcome. Transmits part or all of the effect of the exposure to the outcome.

12. Regression

Continuous Data: Quantitative variables that can any value within a given range, including decimal or fractions. It is measured, not counted.

Binary Data: A categorical variable that can only take two possible values, usually yes/no, true/false, or 0/1.

Ordinal Data: A categorical variable where the categories have a meaningful order/ranking, but the differences between categories are not numerically measured.

Count Data: Numerical data that represents a number of times an event occurs. Non-negative whole numbers and cannot be a fraction or decimal.

Basic Steps in A Regression Analysis:

- Identify your question to be answered
- Understand your data
- Identify the right analytical strategy
- Identify potential covariates (i.e., confounders) to include in the analysis
- Perform regression
- Assess assumptions through diagnostics
- Iterate until the analyses meet the assumption
- Infer conclusions

P-Value: Probability of observing a result greater than this.

95% Confidence Interval: If the analysis was performed 100 times you would expect the results to be within this bound 95 times.

Linear Regression: Regression system used for a continuous data outcome.

Logistic Regression: Regression system used for a binary outcome.

Poisson Regression: Regression system used for a count outcome.

Ordinal Regression: Regression system used for an ordinal Outcome.

Intercept: Expected mean value of the outcome (Y), when all exposures and covariates are held at 0.

Coefficient: Estimated change in the outcome (Y) with a one unit increase in the exposure/covariate (X).

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Error: The unobserved variable, determined as the difference in actual versus estimated data.

Residuals: Difference of actual versus estimated regression points. Used for diagnostics of the regression model.

Assumptions: Diagnostic markers that need to be held to provide valid and reliable results from a regression. Assumptions are based off of the residuals

- Linearity
- Independence of each observation
- Normality
- Constant (equal) variance

Homoscedasticity: Constant variance between each point in a regression residuals.

Heteroscedasticity: Non-constant variance between each point in a regression residuals. This violates regression assumptions.

Leverage: The 'weight' of one point and how this effects the overall regression.

Outlier: An extreme value that is beyond a particular spread of the data. These data points can overly affect regression model results.

Beta: Coefficient for a linear regression model. 0 is the null.

Odd Ratio: Coefficient for a logistic regression model. 1 is the null.

Multicollinearity: When two or more variables are highly correlated.

Variance Inflation Factor (VIF): Measures severity of correlation. How much variance inflation is due to correlation

glm(Y ~ X, data = data, family = 'gaussian') : Basic regression model formula in R. Base R.

plot(model) : Function to assess regression model diagnostics. Base R.

summary(): Summarizes the basic regression model results. Base R.

ci.lin(model) : Summarizes the basic regression model results with 95% confidence intervals. Epi package.

ci.lin(model, Exp = T) : Summarizes the basic regression model results with 95% confidence intervals, to obtain Odds Ratios from a logistic regression. Epi package.

vif(model) : Assesses variance inflation of the model. Car package.

log() : Performs logarithmic transformation of a variable. Base R.

13. Prediction

Prediction: The process of developing and testing mathematical models (equations) that include multiple factors (obtained from descriptive and causal research) that forecast the risk/probability of an undiagnosed condition (diagnostic) or future outcome (prognostic).

Basic steps in a prediction model:

- Define outcome
- Identify potential data collection method and/or data sets
- Determine sample size
- Assess missing data
- Predictor selection
- Assess potential for predictor non-linearity
- Model development
- Internal validation Prediction performance assessment
- External and transportation validation

Parameter: The number of independent pieces of information in a model that are used to estimate and/or forecast another piece of information (i.e., prediction risk score or outcome).

Predictor: A variable that helps predict/forecast an outcome.

Overfitting: When a prediction model learns the intricacies and patterns in a data set too well, and is less generalizable to external data.

Regularization: Method to prevent overfitting by adding a penalty term to the model's loss function.

Optimism Bias: Developed data will perform better than external data.

Internal Validation: Assessment of prediction model through using the same data multiple times either through bootstrapping or 10-fold cross-validation.

Bootstrapping: Repeatedly resampling a single original dataset with replacement to create multiple "bootstrap samples".

10-fold cross-validation: Partitioning data into 10 equal subsets ("folds"). The model is trained 10 times, each time using 9 folds (90%) for training and 1 fold (10%) for validation.

Internal-External Cross-Validation: Internal validation followed by splitting data by cluster to further test the stability of the prediction performance.

Discrimination: How well a model differentiate between individuals with the outcome and individuals without it. Tested through Area Under the Curve (AUC) or c-statistic.

Calibration: The agreement between predicted risks (derived from a model) and observed risks in the data set. Tested through slope with 95% confidence intervals and visually.

R²: Model explanation of overall variance.

Calibration in the Large: How close a model's average predicted risk is to the overall proportion with the event in the data set. This indicates whether the predictions are systematically too low or too high.

TRIPOD AI+: Reporting guideline designed for prediction. <https://www.equator-network.org/reporting-guidelines/tripod-statement/>

External Validation: Testing the prediction algorithm on a new dataset.

Temporal Validation: Testing the prediction model on data from a new time period

Transportation Validation: Testing the prediction model on data from a different country or population.

pmsampsize() : Function to calculate sample size for prediction models. Pmsampsize R package.

roc(y ~ prediction, ci=TRUE) : Area under the curve function. pROC package.

offset() : Function that creates an offset in a model equation. Used in this case to help calculate the calibration in the large. Base R.

mean () : Calculates mean. Base R.

quantile() : Calculates defined quantile. Base R.

colnames() : Defines column names. Base R.

r_square <- 1 - model\$deviance/model\$null.deviance : Function to calculate R². Base R

roc() : Function to calculate area under the curve. pROC R Package.

val.prob.ci.2() : Function to visualize calibration slope from predictions. CalibrationCurves R Package.